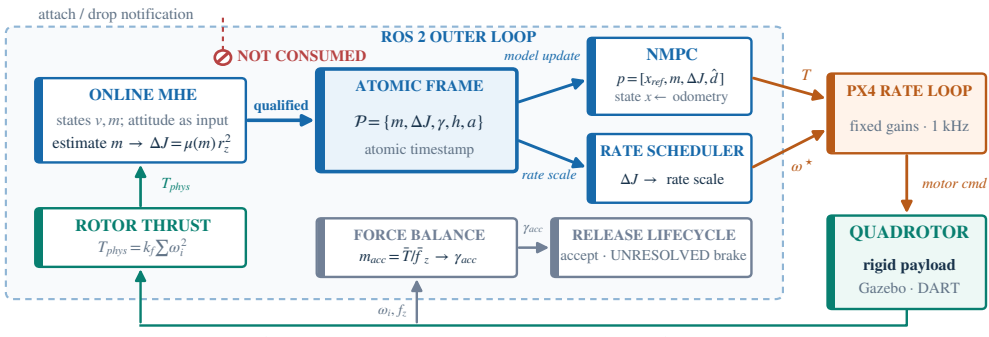
attach / drop notification
ROS 2 OUTER LOOP
NOT CONSUMED
model update
ONLINE MHE
states v, m; attitude as input
estimate $m \rightarrow \Delta J = \mu(m) r_z^2$
qualified
ATOMIC FRAME
$\mathcal{P} = \{m, \Delta J, \gamma, h, a\}$
atomic timestamp
NMPC
$p = [x_{ref}, m, \Delta J, \hat{d}]$
state $x \leftarrow$ odometry
RATE SCHEDULER
$\Delta J \rightarrow$ rate scale
rate scale
T
$\omega^\star$
PX4 RATE LOOP
fixed gains · 1 kHz
motor cmd
T_{phys}
ROTOR THRUST
$T_{phys} = k_f \sum \omega_i^2$
FORCE BALANCE
$m_{acc} = \tilde{T}/\tilde{f}_z \rightarrow \gamma_{acc}$
γ_{acc}
RELEASE LIFECYCLE
accept · UNRESOLVED brake
QUADROTOR
rigid payload
Gazebo · DART
ω_i, f_z
adaptive path: motor speeds ω_i + odometry x only · release evidence: ω_i + IMU f_z